

Numerical Analysis – Lecture 4

...proof. We consider next the case of $\rho(H) < 1$, assuming in addition that H possesses n linearly independent eigenvectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n$, say. Hence $H\mathbf{w}_j = \lambda_j\mathbf{w}_j$, $|\lambda_j| < 1$, $j = 1, 2, \dots, n$. Linear independence means that every $\mathbf{v} \in \mathbb{R}^n$ can be expressed as a linear combination of the eigenvectors. Hence, given $\mathbf{x}_0 \in \mathbb{R}^n$, there exist $\alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{C}$ such that $\mathbf{v}_0 = \mathbf{x}_0 - \mathbf{x}^* = \sum_{j=1}^n \alpha_j \mathbf{w}_j$. Thus,

$$\mathbf{v}_1 = H\mathbf{v}_0 = \sum_{j=1}^n \alpha_j \lambda_j \mathbf{w}_j \quad \text{and, by induction,} \quad \mathbf{v}_m = \sum_{j=1}^n \alpha_j \lambda_j^m \mathbf{w}_j$$

for all $m = 0, 1, \dots$. Since $\rho(H) < 1$, it follows that $\lim_{m \rightarrow \infty} \mathbf{v}_m = \mathbf{0}$, as required. $\square$

The ‘missing’ case Suppose that $\rho(H) < 1$ but that H does not have n linearly independent eigenvalues. This occurs, for example, for the matrix

$$H = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix},$$

where $b \neq 0$ and $|a| < 1$. The eigenvalues of H are both a , but it is an easy exercise to verify that all eigenvectors are necessarily multiples of $\mathbf{e}_1$.

4 QR factorization of matrices

4.1 Scalar products, norms and orthogonality

We revise few definitions. $\mathbb{R}^n$ is the linear space of all real n -tuples. For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ we define the *scalar product*

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{j=1}^n u_j v_j = \mathbf{u}^T \mathbf{v}.$$

The *norm* (a.k.a. the *Euclidean length*) of $\mathbf{u} \in \mathbb{R}^n$ is $\|\mathbf{u}\| := \left(\sum_{j=1}^n u_j^2 \right)^{1/2} = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2}$. Note that $\|\mathbf{u}\| = 0$ iff $\mathbf{u} = \mathbf{0}$.

We say that $\mathbf{u} \in \mathbb{R}^n$ and $\mathbf{v} \in \mathbb{R}^n$ are *orthogonal* to each other if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

The vectors $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_m \in \mathbb{R}^n$ are *orthonormal* if

$$\langle \mathbf{q}_k, \mathbf{q}_\ell \rangle = \begin{cases} 1, & k = \ell, \\ 0, & k \neq \ell, \end{cases} \quad k, \ell = 1, 2, \dots, m.$$

An $n \times n$ real matrix Q is *orthogonal* if all its columns are orthonormal. This is equivalent to $Q^T Q = I$ (I is the *unit matrix*), because $(Q^T Q)_{k,\ell} = \langle \mathbf{q}_k, \mathbf{q}_\ell \rangle$, hence to $Q^{-1} = Q^T$. We conclude that $Q Q^T = Q Q^{-1} = I$ and also the rows of an orthogonal matrix are orthonormal. As a consequence of $1 = \det I = \det(Q Q^T) = \det Q \det Q^T = (\det Q)^2$, we deduce that $\det Q = \pm 1$ and an orthogonal matrix is nonsingular.

Proposition If P, Q are orthogonal then so is PQ .

Proof. Since $P^T P = Q^T Q = I$, we have $(PQ)^T (PQ) = (Q^T P^T)(PQ) = Q^T (P^T P) Q = Q^T Q = I$, hence PQ is orthogonal. $\square$

Proposition Let $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_m \in \mathbb{R}^n$ be orthonormal. Then $m \leq n$.

Proof. Suppose that $m \geq n + 1$ and let Q be the orthogonal matrix whose columns are $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n$. Since Q is nonsingular and $\mathbf{q}_m \neq \mathbf{0}$, there exists a nonzero solution to the linear system $Q\mathbf{a} = \mathbf{q}_m$, hence $\mathbf{q}_m = \sum_{j=1}^n a_j \mathbf{q}_j$. But

$$0 = \langle \mathbf{q}_\ell, \mathbf{q}_m \rangle = \left\langle \mathbf{q}_\ell, \sum_{j=1}^n a_j \mathbf{q}_j \right\rangle = \sum_{j=1}^n a_j \langle \mathbf{q}_\ell, \mathbf{q}_j \rangle = a_j, \quad \ell = 1, 2, \dots, n,$$

hence $\mathbf{a} = \mathbf{0}$, a contradiction. We deduce that $m \leq n$. $\square$

Lemma Let $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_m \in \mathbb{R}^n$ be orthonormal and $m \leq n - 1$. Then there exists $\mathbf{q}_{m+1} \in \mathbb{R}^n$ such that $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_{m+1}$ are orthonormal.

Proof. Let Q be the $n \times m$ matrix whose columns are $\mathbf{q}_1, \dots, \mathbf{q}_m$. Since $\sum_{k=1}^n \sum_{j=1}^m Q_{k,j}^2 = \sum_{j=1}^m \|\mathbf{q}_j\|^2 = m < n$, it follows that $\exists \ell \in \{1, 2, \dots, n\}$ such that $\sum_{j=1}^m Q_{\ell,j}^2 < 1$. We let $\mathbf{w} := \mathbf{e}_\ell - \sum_{j=1}^m \langle \mathbf{q}_j, \mathbf{e}_\ell \rangle \mathbf{q}_j$. Since $Q_{\ell,j} = \langle \mathbf{q}_j, \mathbf{e}_\ell \rangle$, we have $\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w} \rangle = 1 - \sum_{j=1}^m Q_{\ell,j}^2 > 0$ and, by construction, $\mathbf{w}$ is orthogonal to $\mathbf{q}_1, \dots, \mathbf{q}_m$. We set $\mathbf{q}_{m+1} = \mathbf{w}/\|\mathbf{w}\|$. $\square$

4.2 The QR factorization

The QR factorization of an $m \times n$ matrix A has the form $A = QR$, where Q is an $m \times m$ *orthogonal* matrix and R is an $m \times n$ *upper triangular* matrix (i.e., $R_{i,j} = 0$ for $i > j$). We will demonstrate in the sequel that every matrix has a (non-unique) QR factorization.

An application Let $m = n$ and A be nonsingular. We can solve $A\mathbf{x} = \mathbf{b}$ by calculating the QR factorization of A and solving first $Q\mathbf{y} = \mathbf{b}$ (hence $\mathbf{y} = Q^T \mathbf{b}$) and then $R\mathbf{x} = \mathbf{y}$ (a triangular system!).

Interpretation of the QR factorization Let $m \geq n$ and denote the columns of A and Q by $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ and $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_m$ respectively. Since

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \dots \quad \mathbf{q}_m] \begin{bmatrix} R_{1,1} & R_{1,2} & \dots & R_{1,n} \\ 0 & R_{2,2} & & \vdots \\ \vdots & \ddots & \ddots & \\ & & 0 & R_{n,n} \\ \vdots & & & \vdots \\ 0 & \dots & \dots & 0 \end{bmatrix},$$

we have $\mathbf{a}_k = \sum_{j=1}^k R_{j,k} \mathbf{q}_j$, $k = 1, 2, \dots, n$. In other words, Q has the property that each k th column of A can be expressed as a linear combination of the first k columns of Q .